



Properties and Applications of Non-Ferrous Metals 1

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Non-Ferrous Metals

Non-Ferrous Metals are those **NOT** based on Fe:
that is, all the other metals that are not made from iron.

Why do we not make everything out of steel?

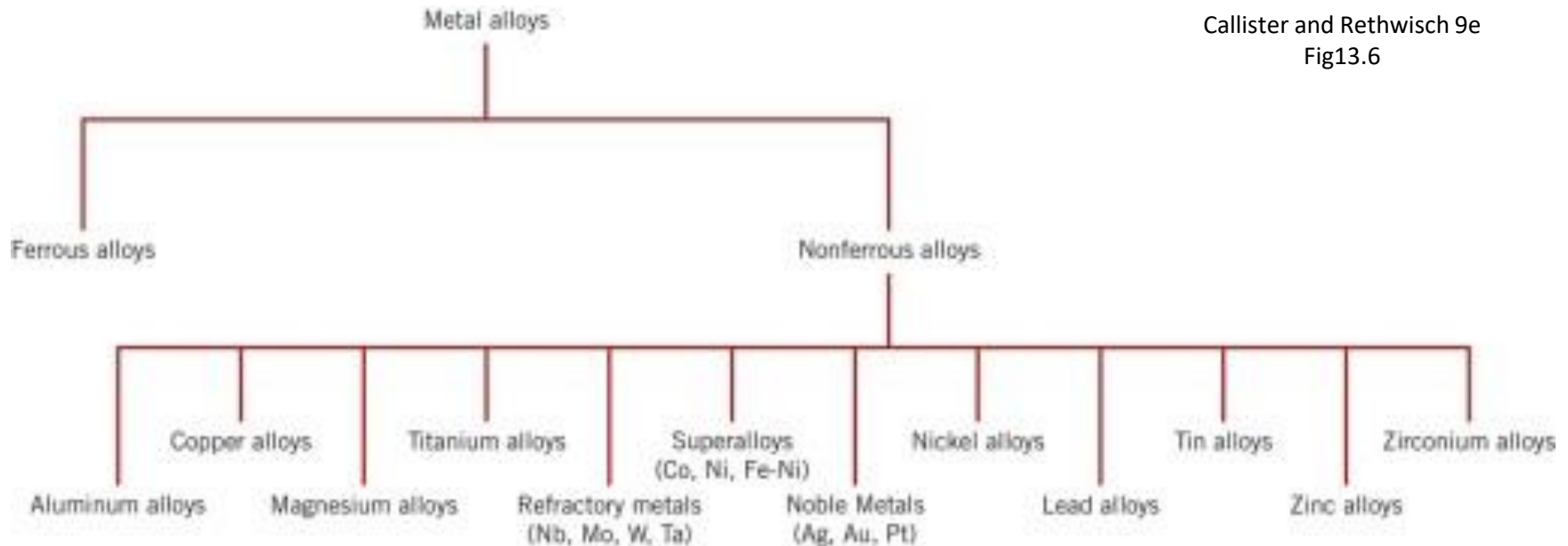
Disadvantages of steel:

- Relatively high density = 7850 kg/m^3 (Al = 2700 kg/m^3)
- Relatively low thermal conductivity = 90 W/mK (Al = 237 W/mK)
- Poor corrosion resistance = oxidation is easy and continuous

Non-Ferrous Metals

Classification of non-ferrous metals:

Callister and Rethwisch 9e
Fig13.6



Some definitions for metals:

- **wrought alloys:** can be shaped by deformation
- **cast alloys:** can only be shaped by casting (often brittle)
- **heat treatable:** different phases formed by heat treatments
e.g. precipitation hardening or martensitic transformation

Non-Ferrous Metals

Comparison of properties:

Alloy system	Examples	Elastic mod GPa	Density kg/m ³	Melting T °C	Notes
Ferrous	Steel	200	7850	1536	Corrosion
Light metals	Al	70	2700	660	Low density Low T _m
	Mg	45	1700	650	
	Ti	110	4500	1660	High specific strength
Transition metals	Cu	120	8900	1083	Good corrosion resistance
	Ni	210	8900	1453	
Refractory metals	W	400	19300	3410	Very high T _m
	Mo	329	10200	2610	
Noble metals	Au	80	19300	1062	No corrosion
	Ag	76	10500	962	
	Pt	170	21400	1770	

Aluminium alloys

Features of aluminium alloys:

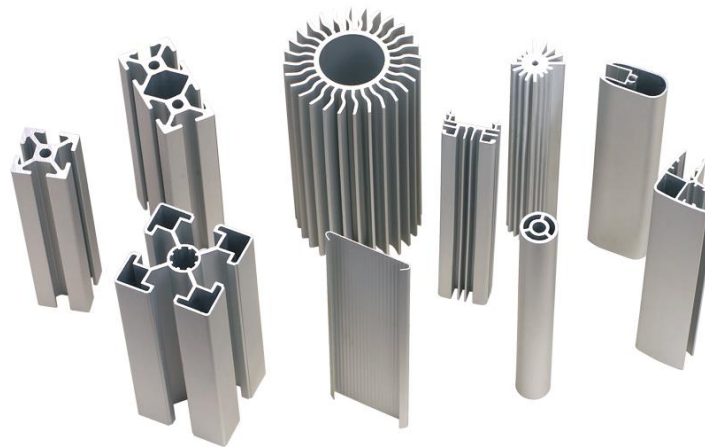


(a)

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Aluminium alloys

Features of aluminium alloys:

- Low density = 2700 kg/m^3 (unlike steel 7850 kg/m^3)
- Low elastic modulus = 70 GPa
- fcc crystal structure = ductile to low temperature
- Low melting point = 660°C (creep $> 100^\circ\text{C}$, grain growth $> 300^\circ\text{C}$)
- Good corrosion resistance (passive oxide)

Very little of any element dissolves in aluminium

So alloys must be carefully controlled to have good properties

If not, impurity phases can cause problems

- Wide range of alloys and properties possible
- High specific strength = strength/density

Aluminium alloys

Classification of aluminium alloys

Each series is characterised by the main alloy element:

Series	Major Alloying Element	Heat Treatable
1XXX	None (commercially pure)	No
2XXX	Copper	Yes
3XXX	Manganese	No
4XXX	Silicon	No (mostly)
5XXX	Magnesium	No
6XXX	Magnesium and Silicon	Yes
7XXX	Zinc	Yes

Suffix	Treatment
-F	As fabricated
-O	Annealed
-HX	Cold worked (strain hardened)
-TX	Solution heat treated, precipitation hardened

The treatment is given by a suffix: e.g. alloy name = 7075-T6

Aluminium alloys

Wrought aluminium alloys – not heat treatable

These alloys are strengthened by deformation (cold work)

Very good corrosion resistance

Alloy	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
1100	Annealed (O)	90	35	35-45
3003	Annealed (O)	110	40	30-40
5052	Work hardened (H32)	230	195	12-18

Applications:

Food – containers, cooking, pressure vessels

Aircraft – fuel pipes and fuel tanks

Aluminium alloys

Wrought aluminium alloys – **heat treatable**

These alloys are strengthened by precipitation hardening
Much higher strengths

Alloy	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
2024	Heat treat (T4)	470	325	20
6061	Heat treat (T4)	240	145	22-25
7075	Heat treat (T6)	570	505	11

Applications:

Aircraft – structures, airframes

Automotive – wheels, frames, pipelines

Aluminium alloys

Cast aluminium alloys – **heat treatable**

Aluminium casting alloys usually contain Si
some is retained as hard silicon particles in the metal
– wear resistant but less ductile

Alloy	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
295.0	Heat treat (T4)	221	110	8.5
356.0	Heat treat (T6)	228	164	3.5

Applications:

Aircraft – wheels, housings and casings

Automotive – engine blocks

Aluminium alloys

Aluminium – Lithium alloys – **heat treatable**

Low density and high strength (new alloys)

Difficult to process because Li is very reactive

Alloy	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
2090	Heat treat and cold work (T83)	455	455	5
8090	Heat treat and cold work (T651)	465	360	brittle

Applications:

Aircraft – structures, cryogenic liquid tanks

Magnesium alloys

Features of magnesium alloys:

- Very low density = 1700 kg/m^3
- Very low elastic modulus = 45 GPa
- hcp crystal structure = difficult to deform at low temperature
- Low melting point = 651°C (creep $> 100^\circ\text{C}$, grain growth $> 300^\circ\text{C}$)
- Good corrosion resistance at room temperature

Difficult to handle – Mg powder will catch fire

Difficult to work without annealing

- High specific strength = strength/density

Magnesium alloys

Properties of magnesium alloys – wrought and cast alloys

Higher specific strengths

Alloy	Condition	σ_{UTS} / MPa	σ_y / MPa	EL / %
Wrought alloys				
M11311	Extruded	262	200	15
M16600	Heat treat (aged)	350	285	11
Cast alloy				
M11916	As cast	230	150	3

Applications:

Aircraft, automotive, computer frames

Titanium alloys

Characteristics of titanium alloys:

Relatively low density (4500 kg/m^3) and high strength (1000 MPa)
= very high specific strength

Relatively low elastic modulus = 107 GPa

hcp crystal structure (bcc at higher temperatures)

High melting temperature = 1668°C

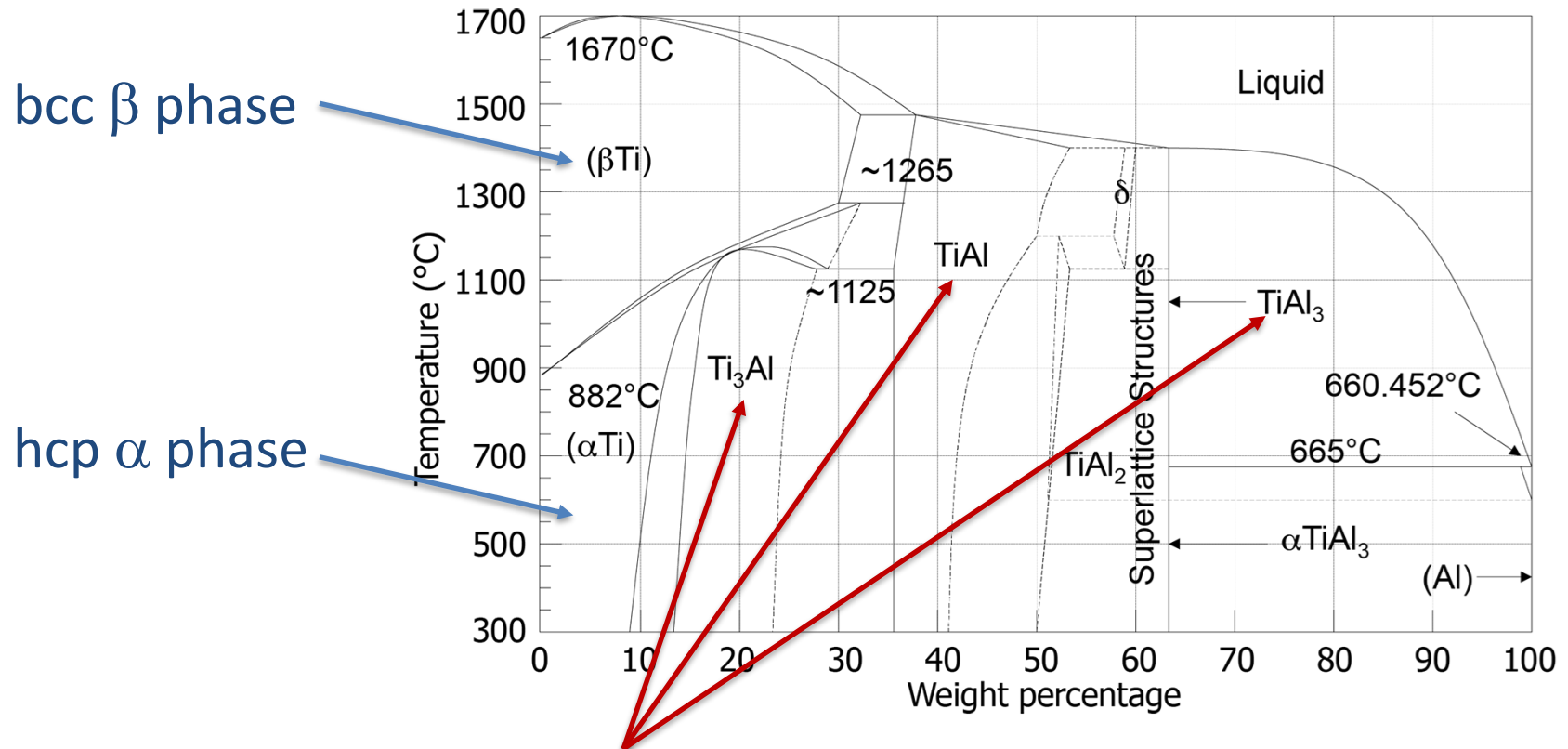
Good corrosion resistance (passive oxide)

- Difficult to process
- Titanium powder will catch fire
- Expensive



Titanium alloys

Titanium phase diagram



Brittle inter-metallic phases

Alloying used to stabilize the β phase to room temperature

Titanium alloys

Properties of titanium alloys

Alloy	Phase	σ_{UTS} / MPa	σ_y / MPa	EL / %
99.5 Ti	Pure Ti	484	414	25
Ti 5Al 2.5Sn	α	826	784	16
Ti 8Al 1Mo 1V	Near α	950	890	15
Ti 6Al 4V	$\alpha - \beta$	947	877	14
Ti 6Al 6V 2Sn	$\alpha - \beta$	1050	985	14
Ti 10V 2Fe 3Al	β	1223	1150	10

Applications:

Aircraft – structures, jet engines, rocket engines

Space – spacecraft propulsion

Medical – joint replacement